

Project Quantum Go: Ising Feature Maps and Photonic Graph Features

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We study Go as a source of features for downstream learning (e.g., winner prediction) using two complementary pathways: (i) a symmetric classical Ising model translated to QUBO for adiabatic/annealing solvers, and (ii) photonic-inspired graph features (Gaussian Boson Sampling, GBS) over a chain-configuration graph (CFG) of the board. We clarify hardware constraints (annealers solve classical Ising/QUBO), present a symmetric reformulation with auxiliary bits to remove color bias, and outline how GBS-derived features capture higher-order group structure. Pending items are explicitly marked as planned work.

OBJECTIVE AND SCOPE

The goal is to derive interpretable energy and graph features of Go positions to feed learning models (e.g., predicting the winner or critical moves). We pursue two lines: a hardware-aligned Ising/QUBO formulation for annealers, and a photonic-style sampler on the CFG of the board to extract subgraph statistics. We did not execute on real D-Wave hardware (access limits), and quantum Hamiltonians with non-diagonal terms were only simulated (PennyLane).

HARDWARE CONSTRAINTS

Adiabatic/quantum annealers natively solve classical Ising/QUBO (diagonal in Z). The initial quantum Hamiltonian with X/Z cross terms cannot be sent directly to an annealer; it was used only for simulation and intuition (time evolution). The practical path is a symmetric classical Ising that can be mapped to QUBO and embedded on annealer topology.

CLASSICAL ISING MODEL FOR GO

Baseline (Atomic) and bias

The Atomic model is quadratic and asymmetric: energy depends on color in a biased way and ignores rule constraints. It provides a coarse energy map but lacks symmetry and stability.

Symmetric model with auxiliaries

We introduce auxiliary bits and penalties to enforce color symmetry and valid states (keep a stone of one color, handle empty). Target energy table (2-body) is enforced via:

- Split spins into positive/negative binaries p_i, n_i with $p_i + n_i \leq 1$.

- Penalize invalid color/vacuum assignments; tune μ for liberties; add constraints for ko/captures (Appendix).
- Match a symmetric energy table across colors (planned: include full table).

From Ising to QUBO

Using the p/n encoding, cubic terms (e.g., $s_0 s_1^2$) are flattened to quadratic with penalties. Linear biases b and couplers Q are produced; coefficients must be scaled to hardware ranges. Embedding onto Pegasus/Zephyr is pending once access is available.

Planned validation

Correlation of energy maps with game outcome, stability of local minima, and comparison against the Atomic baseline will be added when annealing simulations complete.

DATA AND PIPELINE

We process 6,000 professional SGF games via a rules engine (`src/go_game_engine.py`) enforcing ko, superko, captures, and suicide. Each move yields:

- Board states with liberties/captures.
- Classical Ising energies (baseline and symmetric).
- Exports to PNG/GIF through `go_visualization.py`; interactive navigation is summarized in `docs/analisis_interactivo_partidas-go.md`.

PHOTONIC/GBS GRAPH FEATURES

We map the board to a chain-configuration graph (CFG): nodes are groups of stones, edges encode shared

liberties/proximity. A GBS sampler over this graph provides:

- Frequent subgraphs (“super-groups”).
- Cliques (largest solid groups).
- Correlation matrix of nodes (which groups co-occur).
- Sample entropy (ambiguity of the position).
- Temporal similarity between successive positions.

Planned: quantitative tables for representative 9×9 games (mean_subgraph_size, max_clique_size, node_concentration, sample_entropy, graph_similarity).

EXPERIMENTS AND CURRENT STATUS

Simulation (PennyLane)

Time evolution under the quantum Hamiltonian ($I \otimes Z + Z \otimes X + X \otimes Z$) is simulated on small boards (5×5) to visualize amplitude flows; this is illustrative only (not executable on annealers).

QUBO derivation

Code in the notebook translates the symmetric 2-qubit energy table to QUBO via p/n encoding and validates against the target energies. Extension to full board with constraints is planned; coefficient scaling and embedding remain pending.

GBS features

We extract CFG features on selected SGF games; example outputs include clique sizes, entropy trends, and correlation heatmaps. Aggregated metrics versus game outcome are pending.

DISCUSSION

The symmetric Ising removes color bias and is annealer-aligned, but depends on hyperparameters (J, h, μ , penalty weights) and embedding overhead. GBS features capture higher-order structure without explicit rule encoding but are hardware-specific and currently illustrative.

FUTURE WORK

Access an annealer to test the QUBO embedding; integrate energy/GBS features into a predictive model; add an energy tab to the interactive UI; benchmark against classical baselines (e.g., MCTS heuristics); study robustness to noisy SGF records.

CONCLUSIONS

We aligned Go feature extraction with hardware-feasible Ising/QUBO formulations and complementary photonic graph sampling. Pending results (annealer runs and aggregated metrics) will quantify the utility of these features for prediction.

Rule constraints formalization

Constraints encode liberties, captures, and ko as penalties that raise the energy of illegal or unstable patterns. These terms adapt to board size and ko variant; full specification follows the project documentation.

Figures (planned and present)

- Board vs energy map (present): Fig. ??, images at `../data/assets/articulo/Tablero.jpg` and `../data/assets/articulo/Mapa_energia.jpg`.
- Target symmetric energy table (planned): 2-qubit table showing color symmetry after penalties.
- Pipeline schematic (planned): board $\rightarrow$ symmetric Ising $\rightarrow$ QUBO $\rightarrow$ annealer/simulation.
- CFG + GBS output (planned): example sub-graph/clique highlight and correlation heatmap for a 9×9 game.
- Entropy/similarity curves over moves (planned): sample_entropy and graph_similarity vs move number.

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 - [2] A. Lucas, *Ising formulations of many NP problems*, Frontiers in Physics **2**, 5 (2014).
 - [3] Project Quantum Go repository, <https://github.com/ometitlan/Project-Quantum-Go>